

Big Basket-E-commerce-Database-Management-System

Chapter 1

1.1 Introduction:

The Big Basket – Online Grocery Store Management System is a database management system designed to simplify the process of purchasing groceries online. The system maintains information about customers, grocery products, categories, suppliers, orders, payments, inventory, and deliveries in a structured database.

The main purpose of this project is to provide an efficient way to manage all grocery store operations electronically. Customers can browse products, place orders, and make payments, while the administrator can manage products, inventory, suppliers, and customer orders.

project demonstrates the use of Database Management System (DBMS) concepts such as data storage, data retrieval, relationships between tables, primary keys, foreign keys, normalization, and SQL queries. It improves accuracy, reduces manual work, and enables quick access to information.

1.2 Problem Statement:

The **Big Basket** is designed to solve the challenges of managing grocery store data manually. Maintaining customer information, products, inventory, orders, payments, and deliveries without a centralized database can lead to data redundancy, errors, and inefficient operations.

This project provides a structured DBMS solution that stores and manages all grocery-related information in a single database. It ensures accurate data management, faster order processing, efficient inventory tracking, and easy retrieval of information using SQL and relational database concepts.

1.3 Objectives:

The primary objective of this project is to design and develop a database for an online grocery

store that efficiently manages customers, products, orders, payments, and inventory using DBMS concepts.

Key Objectives:

- To maintain customer information securely.
- To organize grocery products into different categories.
- To manage product inventory and stock availability.
- To store supplier information for each product.
- To process and maintain customer orders.
- To record payment details and payment status.
- To manage delivery information for every order.
- To reduce data redundancy through proper database design.
- To ensure data accuracy and consistency using constraints and relationships.
- To generate reports for customers, products, orders, inventory, and sales.

1.4 Motivation of the Project:

- With the rapid growth of online shopping, grocery delivery platforms have become an essential part of daily life. Customers prefer purchasing groceries online because it saves time, offers convenience, and provides a wide variety of products.
- Managing a large number of products, customers, suppliers, and orders manually is time-consuming and prone to errors. A well-designed database helps store and retrieve information efficiently while maintaining data integrity.
- The motivation behind this project is to understand how a real-world online grocery platform works and to apply DBMS concepts such as database design, relationships, normalization, and SQL operations to solve practical business problems. This project also helps in improving database management skills and demonstrates how technology can make grocery shopping faster, more organized, and more efficient. Big basket means 10 minutes delivery.

1.5 Stakeholders:

- **Customers** – Browse grocery products, place orders, and make payments.
- **Administrator (Admin)** – Manages customers, products, categories, inventory, orders, and reports.
- **Suppliers** – Supply grocery products and update product availability.
- **Delivery Personnel** – Deliver customer orders and update delivery status.
- **Store Manager** – Monitors inventory, sales, and overall store operations.

1.6 Business Requirements:

- The system should maintain customer information.
- The system should store and manage grocery product details.
- The system should organize products into categories.
- The system should maintain inventory and stock availability.
- The system should manage supplier information.
- The system should allow customers to place and track orders.
- The system should record payment details and payment status.
- The system should maintain delivery information for each order.
- The system should generate reports on customers, products, orders, inventory, and sales.
- The system should ensure data accuracy, consistency, and security using DBMS constraints.

1.7 Functional Requirements:

- Customer registration and management.
- Admin login and management.
- Add, update, delete, and view grocery products.
- Manage product categories.
- Manage supplier details.
- Maintain inventory and stock levels.
- Place and manage customer orders.
- Record and update payment details.
- Manage delivery information and delivery status.

- Search and view products.
- Generate reports for customers, products, orders, inventory, and sales.

1.8 Non-Functional Requirements:

- **Performance:** The system should retrieve and update data quickly.
- **Reliability:** The database should store data accurately and consistently.
- **Security:** Customer, order, and payment information should be protected from unauthorized access.
- **Scalability:** The database should support an increasing number of customers, products, and orders.
- **Availability:** The system should be available whenever required for data access and management.
- **Maintainability:** The database should be easy to update and maintain.
- **Data Integrity:** The system should ensure data accuracy using primary keys, foreign keys, and constraints.
- **Usability:** The database structure should be simple and easy for administrators to manage.

1.9 Conclusion:

The requirement analysis for the Big Basket provides a clear understanding of the project's objectives, stakeholders, and system requirements. It identifies the key entities such as customers, products, categories, suppliers, inventory, orders, payments, and deliveries that are essential for designing the database.

This analysis serves as the foundation for the next phase of the project, where the ER diagram, relational schema, database tables, relationships, and SQL queries will be developed. By following these requirements, the project aims to create an efficient, reliable, and well-structured database for managing online grocery store operations.